

March 23, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50310808

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 09, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids
Pace Project No.: 50310808

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130
USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50310808

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50310808001	Digester #4	Solid	03/08/22 13:30	03/09/22 08:20

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50310808

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50310808001	Digester #4	EPA 8081	KAV	21	PASI-I
		EPA 8081	KAV	8	PASI-I
		EPA 8082	RID	8	PASI-I
		EPA 8151A	BJW	3	PASI-I
		EPA 6010	DJS	6	PASI-I
		EPA 6010	JPk	8	PASI-I
		EPA 6020	CAW	10	PASI-I
		EPA 7470	ILP	1	PASI-I
		EPA 7471	EAE	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	TMW	55	PASI-I
		EPA 5030/8260	TMW	13	PASI-I
		SM 2540G	BSW	1	PASI-I
		SM 2540G	BSW	1	PASI-I
		SM 2540G	BSW	1	PASI-I
		EPA 9038	SWJ	1	PASI-I
		EPA 9045	SWJ	1	PASI-I
		EPA 351.2	MMS	1	PASI-I
		EPA 353.2	OAS	2	PASI-I
		SM 4500-NH3 G	MMS	1	PASI-I
		SM 4500-Cl-E	OAS	1	PASI-I
		SM 4500-CN-E	SKK	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids								
Analytical Method: EPA 8081 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Aldrin	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	309-00-2	
alpha-BHC	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	319-84-6	
beta-BHC	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	319-85-7	
delta-BHC	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	319-86-8	
gamma-BHC (Lindane)	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	58-89-9	
Chlordane (Technical)	<58700	ug/kg	58700	1	03/21/22 11:20	03/22/22 19:19	57-74-9	
alpha-Chlordane	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	5103-71-9	
gamma-Chlordane	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	5103-74-2	
4,4'-DDD	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	72-54-8	
4,4'-DDE	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	72-55-9	
4,4'-DDT	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	50-29-3	
Dieldrin	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	60-57-1	
Endosulfan I	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	959-98-8	
Endosulfan II	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	33213-65-9	
Endosulfan sulfate	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	1031-07-8	
Endrin	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	72-20-8	
Endrin aldehyde	<5870	ug/kg	5870	1	03/21/22 11:20	03/22/22 19:19	7421-93-4	
Heptachlor	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	76-44-8	
Heptachlor epoxide	<2940	ug/kg	2940	1	03/21/22 11:20	03/22/22 19:19	1024-57-3	
Toxaphene	<58700	ug/kg	58700	1	03/21/22 11:20	03/22/22 19:19	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	84	%.	10-141	1	03/21/22 11:20	03/22/22 19:19	2051-24-3	

8081 GCS Pest RV, TCLP

Analytical Method: EPA 8081 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76

Pace Analytical Services - Indianapolis

gamma-BHC (Lindane)	<0.00025	mg/L	0.00025	1	03/20/22 14:20	03/21/22 18:32	58-89-9	
Chlordane (Technical)	<0.0050	mg/L	0.0050	1	03/20/22 14:20	03/21/22 18:32	57-74-9	
Endrin	<0.00050	mg/L	0.00050	1	03/20/22 14:20	03/21/22 18:32	72-20-8	
Heptachlor	<0.00025	mg/L	0.00025	1	03/20/22 14:20	03/21/22 18:32	76-44-8	
Heptachlor epoxide	<0.00025	mg/L	0.00025	1	03/20/22 14:20	03/21/22 18:32	1024-57-3	
Methoxychlor	<0.0025	mg/L	0.0025	1	03/20/22 14:20	03/21/22 18:32	72-43-5	
Toxaphene	<0.0050	mg/L	0.0050	1	03/20/22 14:20	03/21/22 18:32	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	76	%.	10-102	1	03/20/22 14:20	03/21/22 18:32	2051-24-3	

8082 PCB Solids

Analytical Method: EPA 8082 Preparation Method: EPA 3546

Pace Analytical Services - Indianapolis

PCB-1016 (Aroclor 1016)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	12674-11-2	
PCB-1221 (Aroclor 1221)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	11104-28-2	
PCB-1232 (Aroclor 1232)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	11141-16-5	
PCB-1242 (Aroclor 1242)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	53469-21-9	
PCB-1248 (Aroclor 1248)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	12672-29-6	
PCB-1254 (Aroclor 1254)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	11097-69-1	
PCB-1260 (Aroclor 1260)	<96900	ug/kg	96900	1	03/21/22 14:13	03/22/22 13:50	11096-82-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids								
Analytical Method: EPA 8082 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Surrogates								
Tetrachloro-m-xylene (S)	77	%.	35-106	1	03/21/22 14:13	03/22/22 13:50	877-09-8	
8151A CI Acid Herbicides TCLP								
Analytical Method: EPA 8151A Preparation Method: EPA 8151								
Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76								
Pace Analytical Services - Indianapolis								
2,4-D	<0.0050	mg/L	0.0050	1	03/20/22 14:30	03/21/22 18:53	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	1	03/20/22 14:30	03/21/22 18:53	93-72-1	
Surrogates								
2,4-DCAA (S)	109	%.	10-142	1	03/20/22 14:30	03/21/22 18:53	19719-28-9	
6010 MET ICP								
Analytical Method: EPA 6010 Preparation Method: EPA 3050								
Pace Analytical Services - Indianapolis								
Aluminum	7090	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7429-90-5	
Calcium	38900	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7440-70-2	
Magnesium	6850	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7439-95-4	
Phosphorus	46000	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7723-14-0	N2
Potassium	4700	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7440-09-7	
Sodium	6610	mg/kg	3110	1	03/23/22 05:45	03/23/22 09:56	7440-23-5	
6010 MET ICP, TCLP								
Analytical Method: EPA 6010 Preparation Method: EPA 3010								
Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76								
Pace Analytical Services - Indianapolis								
Arsenic	<0.10	mg/L	0.10	1	03/21/22 07:38	03/22/22 00:25	7440-38-2	
Barium	<5.0	mg/L	5.0	1	03/21/22 07:38	03/22/22 00:25	7440-39-3	
Cadmium	<0.050	mg/L	0.050	1	03/21/22 07:38	03/22/22 00:25	7440-43-9	
Chromium	<0.10	mg/L	0.10	1	03/21/22 07:38	03/22/22 00:25	7440-47-3	
Lead	<0.10	mg/L	0.10	1	03/21/22 07:38	03/22/22 00:25	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	03/21/22 07:38	03/22/22 00:25	7782-49-2	
Silver	<0.10	mg/L	0.10	1	03/21/22 07:38	03/22/22 00:25	7440-22-4	
Zinc	0.59	mg/L	0.20	1	03/21/22 07:38	03/22/22 00:25	7440-66-6	
6020 MET ICPMS								
Analytical Method: EPA 6020 Preparation Method: EPA 3050B								
Pace Analytical Services - Indianapolis								
Arsenic	34.7	mg/kg	1.6	1	03/17/22 12:15	03/18/22 23:02	7440-38-2	
Cadmium	5.4	mg/kg	0.80	1	03/17/22 12:15	03/18/22 23:02	7440-43-9	
Chromium	212	mg/kg	3.2	1	03/17/22 12:15	03/18/22 23:02	7440-47-3	
Copper	1770	mg/kg	16.1	10	03/17/22 12:15	03/18/22 21:50	7440-50-8	
Lead	78.5	mg/kg	8.0	5	03/17/22 12:15	03/18/22 22:19	7439-92-1	
Molybdenum	44.2	mg/kg	1.6	1	03/17/22 12:15	03/18/22 23:02	7439-98-7	N2
Nickel	45.8	mg/kg	0.80	1	03/17/22 12:15	03/18/22 23:02	7440-02-0	
Selenium	9.7	mg/kg	1.6	1	03/17/22 12:15	03/18/22 23:02	7782-49-2	
Silver	7.0	mg/kg	0.80	1	03/17/22 12:15	03/18/22 23:02	7440-22-4	
Zinc	3600	mg/kg	161	20	03/17/22 12:15	03/21/22 18:22	7440-66-6	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP								
Analytical Method: EPA 7470 Preparation Method: EPA 7470								
Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76								
Pace Analytical Services - Indianapolis								
Mercury	<0.0020	mg/L	0.0020	1	03/20/22 18:03	03/21/22 09:32	7439-97-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<13.6	mg/kg	13.6	1	03/21/22 10:16	03/21/22 19:09	7439-97-6	
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Acenaphthene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	83-32-9	
Acenaphthylene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	208-96-8	
Anthracene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	120-12-7	
Benzo(a)anthracene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	56-55-3	
Benzo(a)pyrene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	50-32-8	
Benzo(b)fluoranthene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	205-99-2	
Benzo(g,h,i)perylene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	191-24-2	
Benzo(k)fluoranthene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	207-08-9	
Benzyl alcohol	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59	100-51-6	
4-Bromophenylphenyl ether	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	101-55-3	
Butylbenzylphthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	85-68-7	
4-Chloro-3-methylphenol	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59	59-50-7	
4-Chloroaniline	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59	106-47-8	
bis(2-Chloroethoxy)methane	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	111-91-1	
bis(2-Chloroethyl) ether	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	111-44-4	
bis(2chloro1methylethyl) ether	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	108-60-1	
2-Chloronaphthalene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	91-58-7	
2-Chlorophenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	95-57-8	
4-Chlorophenylphenyl ether	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	7005-72-3	
Chrysene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	218-01-9	
Dibenz(a,h)anthracene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	53-70-3	
Dibenzofuran	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	132-64-9	
3,3'-Dichlorobenzidine	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59	91-94-1	
2,4-Dichlorophenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	120-83-2	
Diethylphthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	84-66-2	
2,4-Dimethylphenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	105-67-9	
Dimethylphthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	131-11-3	
Di-n-butylphthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	84-74-2	
4,6-Dinitro-2-methylphenol	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59	534-52-1	
2,4-Dinitrophenol	<574000	ug/kg	574000	1	03/21/22 11:32	03/21/22 19:59	51-28-5	
2,4-Dinitrotoluene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	121-14-2	
2,6-Dinitrotoluene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	606-20-2	
Di-n-octylphthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	117-84-0	
bis(2-Ethylhexyl)phthalate	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	117-81-7	
Fluoranthene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	206-44-0	

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ANALYTICAL RESULTS

Project: Biosolids

Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Fluorene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	86-73-7	
Hexachloro-1,3-butadiene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	87-68-3	
Hexachlorobenzene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	118-74-1	
Hexachlorocyclopentadiene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	77-47-4	
Hexachloroethane	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	67-72-1	
Indeno(1,2,3-cd)pyrene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	193-39-5	
Isophorone	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	78-59-1	
1-Methylnaphthalene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	90-12-0	
2-Methylnaphthalene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	91-57-6	
2-Methylphenol(o-Cresol)	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	95-48-7	
3&4-Methylphenol(m&p Cresol)	<237000	ug/kg	237000	1	03/21/22 11:32	03/21/22 19:59		
Naphthalene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	91-20-3	
2-Nitroaniline	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	88-74-4	
3-Nitroaniline	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	99-09-2	
4-Nitroaniline	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	100-01-6	
Nitrobenzene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	98-95-3	
2-Nitrophenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	88-75-5	
4-Nitrophenol	<574000	ug/kg	574000	1	03/21/22 11:32	03/21/22 19:59	100-02-7	
N-Nitroso-di-n-propylamine	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	621-64-7	
N-Nitrosodiphenylamine	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	86-30-6	
Pentachlorophenol	<574000	ug/kg	574000	1	03/21/22 11:32	03/21/22 19:59	87-86-5	
Phenanthrene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	85-01-8	
Phenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	108-95-2	P1
Pyrene	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	129-00-0	
2,4,5-Trichlorophenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	95-95-4	
2,4,6-Trichlorophenol	<118000	ug/kg	118000	1	03/21/22 11:32	03/21/22 19:59	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	4	%	32-105	1	03/21/22 11:32	03/21/22 19:59	4165-60-0	H7,S0
Phenol-d5 (S)	34	%	35-114	1	03/21/22 11:32	03/21/22 19:59	4165-62-2	H7,S0
2-Fluorophenol (S)	18	%	33-111	1	03/21/22 11:32	03/21/22 19:59	367-12-4	H7,S0
2,4,6-Tribromophenol (S)	18	%	20-121	1	03/21/22 11:32	03/21/22 19:59	118-79-6	H7,S0
2-Fluorobiphenyl (S)	4	%	35-96	1	03/21/22 11:32	03/21/22 19:59	321-60-8	H7,S0
p-Terphenyl-d14 (S)	14	%	31-145	1	03/21/22 11:32	03/21/22 19:59	1718-51-0	H7,S0

8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76

Pace Analytical Services - Indianapolis

1,4-Dichlorobenzene	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	106-46-7	
2,4-Dinitrotoluene	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	121-14-2	
Hexachloro-1,3-butadiene	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	87-68-3	
Hexachlorobenzene	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	118-74-1	
Hexachloroethane	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	67-72-1	
2-Methylphenol(o-Cresol)	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.20	mg/L	0.20	1	03/21/22 10:20	03/21/22 17:51		
Nitrobenzene	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	98-95-3	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
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8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 03/16/22 18:45 Initial pH: 7.81; Final pH: 4.76

Pace Analytical Services - Indianapolis

Pentachlorophenol	<0.50	mg/L	0.50	1	03/21/22 10:20	03/21/22 17:51	87-86-5	
Pyridine	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	110-86-1	L2
2,4,5-Trichlorophenol	<0.50	mg/L	0.50	1	03/21/22 10:20	03/21/22 17:51	95-95-4	
2,4,6-Trichlorophenol	<0.10	mg/L	0.10	1	03/21/22 10:20	03/21/22 17:51	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	67	%	40-115	1	03/21/22 10:20	03/21/22 17:51	4165-60-0	
2-Fluorobiphenyl (S)	54	%	35-102	1	03/21/22 10:20	03/21/22 17:51	321-60-8	
p-Terphenyl-d14 (S)	70	%	42-156	1	03/21/22 10:20	03/21/22 17:51	1718-51-0	
Phenol-d5 (S)	21	%	15-48	1	03/21/22 10:20	03/21/22 17:51	4165-62-2	
2-Fluorophenol (S)	31	%	21-74	1	03/21/22 10:20	03/21/22 17:51	367-12-4	
2,4,6-Tribromophenol (S)	72	%	47-127	1	03/21/22 10:20	03/21/22 17:51	118-79-6	

8260 MSV 5030 Low Level

Analytical Method: EPA 8260

Pace Analytical Services - Indianapolis

Acetone	7.2	mg/kg	6.3	1		03/21/22 17:06	67-64-1	
Acrolein	<6.3	mg/kg	6.3	1		03/21/22 17:06	107-02-8	
Acrylonitrile	<6.3	mg/kg	6.3	1		03/21/22 17:06	107-13-1	
Benzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	71-43-2	
Bromodichloromethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-27-4	
Bromoform	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-25-2	
Bromomethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	74-83-9	
2-Butanone (MEK)	<1.6	mg/kg	1.6	1		03/21/22 17:06	78-93-3	
Carbon disulfide	<0.63	mg/kg	0.63	1		03/21/22 17:06	75-15-0	
Carbon tetrachloride	<0.32	mg/kg	0.32	1		03/21/22 17:06	56-23-5	
Chlorobenzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	108-90-7	
Chloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-00-3	
Chloroform	<0.32	mg/kg	0.32	1		03/21/22 17:06	67-66-3	
Chloromethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	74-87-3	
1,2-Dibromo-3-chloropropane	<0.63	mg/kg	0.63	1		03/21/22 17:06	96-12-8	
Dibromochloromethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	124-48-1	
1,2-Dibromoethane (EDB)	<0.32	mg/kg	0.32	1		03/21/22 17:06	106-93-4	
Dibromomethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	74-95-3	
1,2-Dichlorobenzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	95-50-1	
1,3-Dichlorobenzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	541-73-1	
1,4-Dichlorobenzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	106-46-7	
trans-1,4-Dichloro-2-butene	<6.3	mg/kg	6.3	1		03/21/22 17:06	110-57-6	
Dichlorodifluoromethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-71-8	
1,1-Dichloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-34-3	
1,2-Dichloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	107-06-2	
1,1-Dichloroethene	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-35-4	
cis-1,2-Dichloroethene	<0.32	mg/kg	0.32	1		03/21/22 17:06	156-59-2	
trans-1,2-Dichloroethene	<0.32	mg/kg	0.32	1		03/21/22 17:06	156-60-5	
1,2-Dichloropropane	<0.32	mg/kg	0.32	1		03/21/22 17:06	78-87-5	
cis-1,3-Dichloropropene	<0.32	mg/kg	0.32	1		03/21/22 17:06	10061-01-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis						
trans-1,3-Dichloropropene	<0.32	mg/kg	0.32	1		03/21/22 17:06	10061-02-6	
Ethylbenzene	<0.32	mg/kg	0.32	1		03/21/22 17:06	100-41-4	
Ethyl methacrylate	<6.3	mg/kg	6.3	1		03/21/22 17:06	97-63-2	
2-Hexanone	<6.3	mg/kg	6.3	1		03/21/22 17:06	591-78-6	
Iodomethane	<6.3	mg/kg	6.3	1		03/21/22 17:06	74-88-4	
Methylene Chloride	<1.3	mg/kg	1.3	1		03/21/22 17:06	75-09-2	
Methyl methacrylate	<6.3	mg/kg	6.3	1		03/21/22 17:06	80-62-6	
4-Methyl-2-pentanone (MIBK)	<1.6	mg/kg	1.6	1		03/21/22 17:06	108-10-1	
Methyl-tert-butyl ether	<0.32	mg/kg	0.32	1		03/21/22 17:06	1634-04-4	
Styrene	<0.32	mg/kg	0.32	1		03/21/22 17:06	100-42-5	
1,1,1,2-Tetrachloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	630-20-6	
1,1,2,2-Tetrachloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	79-34-5	
Tetrachloroethene	<0.32	mg/kg	0.32	1		03/21/22 17:06	127-18-4	
Toluene	<0.32	mg/kg	0.32	1		03/21/22 17:06	108-88-3	
1,1,1-Trichloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	71-55-6	
1,1,2-Trichloroethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	79-00-5	
Trichloroethene	<0.32	mg/kg	0.32	1		03/21/22 17:06	79-01-6	
Trichlorofluoromethane	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-69-4	
1,2,3-Trichloropropane	<0.32	mg/kg	0.32	1		03/21/22 17:06	96-18-4	
Vinyl acetate	<6.3	mg/kg	6.3	1		03/21/22 17:06	108-05-4	
Vinyl chloride	<0.32	mg/kg	0.32	1		03/21/22 17:06	75-01-4	
Xylene (Total)	<0.63	mg/kg	0.63	1		03/21/22 17:06	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	107	%.	73-132	1		03/21/22 17:06	1868-53-7	
Toluene-d8 (S)	106	%.	66-148	1		03/21/22 17:06	2037-26-5	
4-Bromofluorobenzene (S)	107	%.	40-149	1		03/21/22 17:06	460-00-4	

8260 MSV TCLP

Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 03/16/22 18:45

Pace Analytical Services - Indianapolis

Benzene	<0.050	mg/L	0.050	1		03/21/22 14:33	71-43-2	
2-Butanone (MEK)	<1.0	mg/L	1.0	1		03/21/22 14:33	78-93-3	
Carbon tetrachloride	<0.050	mg/L	0.050	1		03/21/22 14:33	56-23-5	
Chlorobenzene	<0.050	mg/L	0.050	1		03/21/22 14:33	108-90-7	
Chloroform	<0.050	mg/L	0.050	1		03/21/22 14:33	67-66-3	
1,2-Dichloroethane	<0.050	mg/L	0.050	1		03/21/22 14:33	107-06-2	
1,1-Dichloroethene	<0.050	mg/L	0.050	1		03/21/22 14:33	75-35-4	
Tetrachloroethene	<0.050	mg/L	0.050	1		03/21/22 14:33	127-18-4	
Trichloroethene	<0.050	mg/L	0.050	1		03/21/22 14:33	79-01-6	
Vinyl chloride	<0.020	mg/L	0.020	1		03/21/22 14:33	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	108	%.	78-117	1		03/21/22 14:33	460-00-4	
Dibromofluoromethane (S)	106	%.	78-120	1		03/21/22 14:33	1868-53-7	
Toluene-d8 (S)	100	%.	77-118	1		03/21/22 14:33	2037-26-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50310808

Sample: Digester #4 **Lab ID: 50310808001** Collected: 03/08/22 13:30 Received: 03/09/22 08:20 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Percent Moisture	98.5	%	0.10	1		03/10/22 15:48		N2
2540G Total Percent Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Solids	1.6	%	0.10	1		03/10/22 15:48		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Volatile Solids	58.3	%	0.10	1		03/10/22 15:48		N2
9038 Sulfate Soil								
Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis								
Sulfate	<6450	mg/kg	6450	1	03/10/22 14:00	03/11/22 10:28	14808-79-8	N2
9045 pH Soil								
Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	8.0	Std. Units	0.10	1		03/18/22 14:34		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis								
Nitrogen, Kjeldahl, Total	98900	mg/kg	6360	2	03/21/22 12:41	03/22/22 17:51	7727-37-9	N2
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<3200	mg/kg	3200	10	03/10/22 12:38	03/10/22 14:41		D3,N2
Nitrogen, Nitrate	<3200	mg/kg	3200	10	03/10/22 12:38	03/10/22 14:41	14797-55-8	D3,N2
4500 Ammonia Soil, Distilled								
Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis								
Nitrogen, Ammonia	43300	mg/kg	3110	1	03/14/22 10:15	03/14/22 13:32	7664-41-7	N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis								
Chloride	<160000	mg/kg	160000	25	03/19/22 09:00	03/20/22 12:36	16887-00-6	D3,N2
4500CNE Cyanide, Soil								
Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	<31.8	mg/kg	31.8	1	03/22/22 11:11	03/22/22 18:03	57-12-5	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667101	Analysis Method: EPA 7470
QC Batch Method: EPA 7470	Analysis Description: 7470 Mercury TCLP
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3071705 Matrix: Water
Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	03/21/22 09:22	

LABORATORY CONTROL SAMPLE: 3071706

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0050	99	80-120	

MATRIX SPIKE SAMPLE: 3071707

Parameter	Units	50310594001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	98	75-125	

MATRIX SPIKE SAMPLE: 3071708

Parameter	Units	50310808001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.015	102	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3071709 3071710

Parameter	Units	50310825006 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	ND	0.015	0.015	0.015	0.015	102	100	75-125	2	20	

MATRIX SPIKE SAMPLE: 3071711

Parameter	Units	50311153001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	98	75-125	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

MATRIX SPIKE SAMPLE:		3071712					
		50311187001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	100	75-125	

MATRIX SPIKE SAMPLE:		3071713					
		50311191001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	102	75-125	

MATRIX SPIKE SAMPLE:		3071714					
		50311192001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	99	75-125	

MATRIX SPIKE SAMPLE:		3071715					
		50311195001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	100	75-125	

MATRIX SPIKE SAMPLE:		3071716					
		50311402002	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	100	75-125	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667251	Analysis Method: EPA 7471
QC Batch Method: EPA 7471	Analysis Description: 7471 Mercury
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3072933 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	03/21/22 18:57	

LABORATORY CONTROL SAMPLE: 3072934

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.47	94	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3072935 3072936

Parameter	Units	50310825002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.61	0.6	0.62	0.63	95	97	75-125	1	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 667371

Analysis Method: EPA 6010

QC Batch Method: EPA 3050

Analysis Description: 6010 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073302

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<46.1	46.1	03/23/22 09:54	
Calcium	mg/kg	<46.1	46.1	03/23/22 09:54	
Magnesium	mg/kg	<46.1	46.1	03/23/22 09:54	
Phosphorus	mg/kg	ND	46.1	03/23/22 09:54	N2
Potassium	mg/kg	<46.1	46.1	03/23/22 09:54	
Sodium	mg/kg	<46.1	46.1	03/23/22 09:54	

LABORATORY CONTROL SAMPLE: 3073303

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	485	455	94	80-120	
Calcium	mg/kg	485	476	98	80-120	
Magnesium	mg/kg	485	443	91	80-120	
Phosphorus	mg/kg	48.5	48.5	100	80-120	N2
Potassium	mg/kg	485	465	96	80-120	
Sodium	mg/kg	485	470	97	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073304 3073305

Parameter	Units	50310828001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	9010	539	603	11800	11100	515	349	75-125	6	20	P6
Calcium	mg/kg	2480	539	603	3550	3240	198	127	75-125	9	20	P6
Magnesium	mg/kg	2500	539	603	3090	3090	110	98	75-125	0	20	
Phosphorus	mg/kg	401	53.9	60.3	468	482	124	134	75-125	3	20	N2,P6
Potassium	mg/kg	1140	539	603	1950	1960	150	136	75-125	0	20	M3
Sodium	mg/kg	115	539	603	610	671	92	92	75-125	10	20	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073306 3073307

Parameter	Units	50310828009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	7210	515	556	8910	6630	331	-104	75-125	29	20	P6,R1
Calcium	mg/kg	47800	515	556	65800	64300	3490	2960	75-125	2	20	E,P6
Magnesium	mg/kg	23900	515	556	43200	8620	3740	-2750	75-125	133	20	E,P6, R1

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073306 3073307												
Parameter	Units	50310828009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phosphorus	mg/kg	531	51.5	55.6	319	392	-412	-249	75-125	21	20	N2,P6, R1
Potassium	mg/kg	1030	515	556	2200	1600	227	103	75-125	31	20	M3,R1
Sodium	mg/kg	185	515	556	737	667	107	87	75-125	10	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 667082

QC Batch Method: EPA 3010

Analysis Method: EPA 6010

Analysis Description: 6010 MET TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3071639

Matrix: Water

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	03/22/22 00:22	
Barium	mg/L	<0.50	0.50	03/22/22 00:22	
Cadmium	mg/L	<0.0050	0.0050	03/22/22 00:22	
Chromium	mg/L	<0.010	0.010	03/22/22 00:22	
Lead	mg/L	<0.010	0.010	03/22/22 00:22	
Selenium	mg/L	<0.010	0.010	03/22/22 00:22	
Silver	mg/L	<0.010	0.010	03/22/22 00:22	
Zinc	mg/L	<0.020	0.020	03/22/22 00:22	

LABORATORY CONTROL SAMPLE: 3071640

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	1.1	105	80-120	
Barium	mg/L	1	1.0	102	80-120	
Cadmium	mg/L	1	1.0	101	80-120	
Chromium	mg/L	1	1.0	102	80-120	
Lead	mg/L	1	0.96	96	80-120	
Selenium	mg/L	1	1.0	104	80-120	
Silver	mg/L	0.5	0.49	98	80-120	
Zinc	mg/L	1	0.99	99	80-120	

MATRIX SPIKE SAMPLE: 3071641

Parameter	Units	50310808001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.6	106	50-150	
Barium	mg/L	<5.0	10	10.2	102	50-150	
Cadmium	mg/L	<0.050	10	10.0	100	50-150	
Chromium	mg/L	<0.10	10	10.0	100	50-150	
Lead	mg/L	<0.10	10	9.3	93	50-150	
Selenium	mg/L	<0.10	10	10.4	104	50-150	
Silver	mg/L	<0.10	5	4.9	97	50-150	
Zinc	mg/L	0.59	10	10.5	99	50-150	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3071642 3071643											
Parameter	Units	50310825006 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.5	10.5	105	105	50-150	0	20
Barium	mg/L	ND	10	10	10.2	10.3	100	101	50-150	1	20
Cadmium	mg/L	ND	10	10	10.0	10.1	100	101	50-150	0	20
Chromium	mg/L	ND	10	10	9.9	9.9	99	99	50-150	1	20
Lead	mg/L	ND	10	10	9.3	9.2	92	92	50-150	0	20
Selenium	mg/L	ND	10	10	10.4	10.4	104	104	50-150	0	20
Silver	mg/L	ND	5	5	4.8	4.8	96	96	50-150	0	20
Zinc	mg/L	ND	10	10	9.8	9.9	98	98	50-150	0	20

MATRIX SPIKE SAMPLE: 3071644							
Parameter	Units	50311153001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.7	107	50-150	
Barium	mg/L	ND	10	10.2	101	50-150	
Cadmium	mg/L	ND	10	10.2	101	50-150	
Chromium	mg/L	27.1	10	39.5	124	50-150	
Lead	mg/L	0.22	10	9.4	92	50-150	
Selenium	mg/L	0.080J	10	10.7	106	50-150	
Silver	mg/L	ND	5	4.9	97	50-150	
Zinc	mg/L	11400 ug/L	10	22.2	108	50-150	

MATRIX SPIKE SAMPLE: 3071645							
Parameter	Units	50311187001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.8	107	50-150	
Barium	mg/L	ND	10	10.3	103	50-150	
Cadmium	mg/L	ND	10	10.2	102	50-150	
Chromium	mg/L	3.2	10	13.5	103	50-150	
Lead	mg/L	ND	10	9.1	91	50-150	
Selenium	mg/L	0.29	10	10.8	105	50-150	
Silver	mg/L	ND	5	5.0	99	50-150	
Zinc	mg/L	1280 ug/L	10	11.3	100	50-150	

MATRIX SPIKE SAMPLE: 3071646							
Parameter	Units	50311191001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.4	104	50-150	
Barium	mg/L	ND	10	12.6	102	50-150	
Cadmium	mg/L	0.27	10	10.5	102	50-150	
Chromium	mg/L	ND	10	10	99	50-150	
Lead	mg/L	ND	10	9.2	92	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

MATRIX SPIKE SAMPLE:		3071646					
Parameter	Units	50311191001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Selenium	mg/L	ND	10	10.5	104	50-150	
Silver	mg/L	ND	5	4.9	98	50-150	
Zinc	mg/L	113000 ug/L	10	129	157	50-150	P6

MATRIX SPIKE SAMPLE:		3071647					
Parameter	Units	50311192001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.3	103	50-150	
Barium	mg/L	10.4	10	21.1	107	50-150	
Cadmium	mg/L	ND	10	10.1	101	50-150	
Chromium	mg/L	ND	10	10	100	50-150	
Lead	mg/L	ND	10	9.1	91	50-150	
Selenium	mg/L	0.10	10	10.3	102	50-150	
Silver	mg/L	ND	5	4.9	97	50-150	
Zinc	mg/L	ND	10	9.9	98	50-150	

MATRIX SPIKE SAMPLE:		3071648					
Parameter	Units	50311195001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.8	108	50-150	
Barium	mg/L	ND	10	11.1	102	50-150	
Cadmium	mg/L	ND	10	10.3	103	50-150	
Chromium	mg/L	ND	10	10.1	101	50-150	
Lead	mg/L	ND	10	9.4	94	50-150	
Selenium	mg/L	ND	10	10.7	106	50-150	
Silver	mg/L	ND	5	4.3	86	50-150	
Zinc	mg/L	2230 ug/L	10	12.4	102	50-150	

MATRIX SPIKE SAMPLE:		3071649					
Parameter	Units	50311402002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.8	107	50-150	
Barium	mg/L	ND	10	10.6	102	50-150	
Cadmium	mg/L	ND	10	10.4	104	50-150	
Chromium	mg/L	ND	10	10	100	50-150	
Lead	mg/L	ND	10	9.4	93	50-150	
Selenium	mg/L	ND	10	10.7	106	50-150	
Silver	mg/L	ND	5	4.3	85	50-150	
Zinc	mg/L	ND	10	10.1	100	50-150	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 666526

QC Batch Method: EPA 3050B

Analysis Method: EPA 6020

Analysis Description: 6020 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3068980

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	03/18/22 21:07	
Cadmium	mg/kg	<0.050	0.050	03/18/22 21:07	
Chromium	mg/kg	<0.20	0.20	03/18/22 21:07	
Copper	mg/kg	<0.10	0.10	03/18/22 21:07	
Lead	mg/kg	<0.10	0.10	03/18/22 21:07	
Molybdenum	mg/kg	<0.10	0.10	03/18/22 21:07	N2
Nickel	mg/kg	<0.050	0.050	03/18/22 21:07	
Selenium	mg/kg	<0.10	0.10	03/18/22 21:07	
Silver	mg/kg	<0.050	0.050	03/18/22 21:07	
Zinc	mg/kg	<0.50	0.50	03/18/22 21:07	

LABORATORY CONTROL SAMPLE: 3068981

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.7	92	80-120	
Cadmium	mg/kg	4	3.7	93	80-120	
Chromium	mg/kg	4	4.0	101	80-120	
Copper	mg/kg	4	4.0	101	80-120	
Lead	mg/kg	4	4.1	101	80-120	
Molybdenum	mg/kg	4	4.1	102	80-120	N2
Nickel	mg/kg	4	3.9	99	80-120	
Selenium	mg/kg	4	3.8	95	80-120	
Silver	mg/kg	4	4.1	101	80-120	
Zinc	mg/kg	4	3.7	92	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3068982 3068983

Parameter	Units	5290936001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	ND	8.1	8.2	7.9	7.9	97	96	75-125	0	20	
Cadmium	mg/kg	ND	8.1	8.2	7.8	7.8	94	95	75-125	1	20	
Chromium	mg/kg	ND	8.1	8.2	8.4	8.4	102	102	75-125	0	20	
Copper	mg/kg	ND	8.1	8.2	8.3	8.3	101	100	75-125	1	20	
Lead	mg/kg	ND	8.1	8.2	8.3	8.4	101	103	75-125	2	20	
Molybdenum	mg/kg	ND	8.1	8.2	8.4	8.5	102	103	75-125	2	20	N2
Nickel	mg/kg	0.20	8.1	8.2	8.1	8.1	97	96	75-125	0	20	
Selenium	mg/kg	ND	8.1	8.2	8.7	8.7	105	105	75-125	0	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3068982 3068983												
Parameter	Units	5290936001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Silver	mg/kg	ND	8.1	8.2	8.3	8.4	101	102	75-125	1	20	
Zinc	mg/kg	ND	8.1	8.2	7.9	8.0	92	92	75-125	1	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667532	Analysis Method: EPA 8260
QC Batch Method: EPA 8260	Analysis Description: 8260 MSV 5030 Low
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073804 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	03/21/22 11:31	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
2-Butanone (MEK)	mg/kg	<0.025	0.025	03/21/22 11:31	
2-Hexanone	mg/kg	<0.10	0.10	03/21/22 11:31	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	03/21/22 11:31	
Acetone	mg/kg	<0.10	0.10	03/21/22 11:31	
Acrolein	mg/kg	<0.10	0.10	03/21/22 11:31	
Acrylonitrile	mg/kg	<0.10	0.10	03/21/22 11:31	
Benzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Bromodichloromethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Bromoform	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Bromomethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Carbon disulfide	mg/kg	<0.010	0.010	03/21/22 11:31	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Chlorobenzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Chloroethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Chloroform	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Chloromethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Dibromochloromethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Dibromomethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Ethyl methacrylate	mg/kg	<0.10	0.10	03/21/22 11:31	
Ethylbenzene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Iodomethane	mg/kg	<0.10	0.10	03/21/22 11:31	
Methyl methacrylate	mg/kg	<0.10	0.10	03/21/22 11:31	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	03/21/22 11:31	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

METHOD BLANK: 3073804

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	03/21/22 11:31	
Styrene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Tetrachloroethene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Toluene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	03/21/22 11:31	
Trichloroethene	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Vinyl acetate	mg/kg	<0.10	0.10	03/21/22 11:31	
Vinyl chloride	mg/kg	<0.0050	0.0050	03/21/22 11:31	
Xylene (Total)	mg/kg	<0.010	0.010	03/21/22 11:31	
4-Bromofluorobenzene (S)	%	113	40-149	03/21/22 11:31	
Dibromofluoromethane (S)	%	105	73-132	03/21/22 11:31	1d
Toluene-d8 (S)	%	100	66-148	03/21/22 11:31	

LABORATORY CONTROL SAMPLE: 3073805

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1-Trichloroethane	mg/kg	0.05	0.048	97	68-129	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.050	99	67-137	
1,1-Dichloroethene	mg/kg	0.05	0.042	85	53-135	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.048	97	68-125	
1,2-Dichloroethane	mg/kg	0.05	0.052	104	69-128	
1,2-Dichloropropane	mg/kg	0.05	0.051	102	70-130	
Benzene	mg/kg	0.05	0.049	97	69-125	
Chlorobenzene	mg/kg	0.05	0.047	93	66-121	
Chloroform	mg/kg	0.05	0.048	96	66-123	
cis-1,2-Dichloroethene	mg/kg	0.05	0.047	94	67-122	
Ethylbenzene	mg/kg	0.05	0.048	96	57-126	
Methyl-tert-butyl ether	mg/kg	0.05	0.049	97	66-136	
Tetrachloroethene	mg/kg	0.05	0.046	92	61-123	
Toluene	mg/kg	0.05	0.047	94	67-128	
trans-1,2-Dichloroethene	mg/kg	0.05	0.046	92	61-127	
Trichloroethene	mg/kg	0.05	0.049	98	64-122	
Vinyl chloride	mg/kg	0.05	0.035	71	42-148	
Xylene (Total)	mg/kg	0.15	0.14	95	62-126	
4-Bromofluorobenzene (S)	%			107	40-149	
Dibromofluoromethane (S)	%			105	73-132	
Toluene-d8 (S)	%			101	66-148	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch:	667363	Analysis Method:	EPA 5030/8260
QC Batch Method:	EPA 5030/8260	Analysis Description:	8260 MSV TCLP
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073284 Matrix: Water

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	03/21/22 11:31	
1,2-Dichloroethane	mg/L	<0.050	0.050	03/21/22 11:31	
2-Butanone (MEK)	mg/L	<1.0	1.0	03/21/22 11:31	
Benzene	mg/L	<0.050	0.050	03/21/22 11:31	
Carbon tetrachloride	mg/L	<0.050	0.050	03/21/22 11:31	
Chlorobenzene	mg/L	<0.050	0.050	03/21/22 11:31	
Chloroform	mg/L	<0.050	0.050	03/21/22 11:31	
Tetrachloroethene	mg/L	<0.050	0.050	03/21/22 11:31	
Trichloroethene	mg/L	<0.050	0.050	03/21/22 11:31	
Vinyl chloride	mg/L	<0.020	0.020	03/21/22 11:31	
4-Bromofluorobenzene (S)	%	113	78-117	03/21/22 11:31	1d
Dibromofluoromethane (S)	%	105	78-120	03/21/22 11:31	
Toluene-d8 (S)	%	100	77-118	03/21/22 11:31	

LABORATORY CONTROL SAMPLE: 3073285

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.42	85	67-136	
1,2-Dichloroethane	mg/L	0.5	0.52	104	69-135	
2-Butanone (MEK)	mg/L	2.5	2.9	114	56-164	
Benzene	mg/L	0.5	0.49	97	77-128	
Carbon tetrachloride	mg/L	0.5	0.51	101	61-139	
Chlorobenzene	mg/L	0.5	0.47	93	76-124	
Chloroform	mg/L	0.5	0.48	96	77-120	
Tetrachloroethene	mg/L	0.5	0.46	92	70-124	
Trichloroethene	mg/L	0.5	0.49	98	75-130	
Vinyl chloride	mg/L	0.5	0.35	71	51-140	
4-Bromofluorobenzene (S)	%			107	78-117	
Dibromofluoromethane (S)	%			105	78-120	
Toluene-d8 (S)	%			101	77-118	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667291	Analysis Method: EPA 8081
QC Batch Method: EPA 3546	Analysis Description: 8081 GCS Pesticides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073072 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<4.9	4.9	03/22/22 12:39	
4,4'-DDE	ug/kg	<4.9	4.9	03/22/22 12:39	
4,4'-DDT	ug/kg	<4.9	4.9	03/22/22 12:39	
Aldrin	ug/kg	<2.4	2.4	03/22/22 12:39	
alpha-BHC	ug/kg	<2.4	2.4	03/22/22 12:39	
alpha-Chlordane	ug/kg	<2.4	2.4	03/22/22 12:39	
beta-BHC	ug/kg	<2.4	2.4	03/22/22 12:39	
Chlordane (Technical)	ug/kg	<48.8	48.8	03/22/22 12:39	
delta-BHC	ug/kg	<2.4	2.4	03/22/22 12:39	
Dieldrin	ug/kg	<4.9	4.9	03/22/22 12:39	
Endosulfan I	ug/kg	<2.4	2.4	03/22/22 12:39	
Endosulfan II	ug/kg	<4.9	4.9	03/22/22 12:39	
Endosulfan sulfate	ug/kg	<4.9	4.9	03/22/22 12:39	
Endrin	ug/kg	<4.9	4.9	03/22/22 12:39	
Endrin aldehyde	ug/kg	<4.9	4.9	03/22/22 12:39	
gamma-BHC (Lindane)	ug/kg	<2.4	2.4	03/22/22 12:39	
gamma-Chlordane	ug/kg	<2.4	2.4	03/22/22 12:39	
Heptachlor	ug/kg	<2.4	2.4	03/22/22 12:39	
Heptachlor epoxide	ug/kg	<2.4	2.4	03/22/22 12:39	
Toxaphene	ug/kg	<48.8	48.8	03/22/22 12:39	
Decachlorobiphenyl (S)	%	81	10-141	03/22/22 12:39	

LABORATORY CONTROL SAMPLE: 3073073

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	19.7	27.3	138	55-163	
4,4'-DDE	ug/kg	19.7	26.4	134	55-161	
4,4'-DDT	ug/kg	19.7	25.5	129	43-183	
Aldrin	ug/kg	9.9	12.9	131	50-155	
alpha-BHC	ug/kg	9.9	13.1	132	50-152	
alpha-Chlordane	ug/kg	9.9	12.7	129	53-148	
beta-BHC	ug/kg	9.9	13.9	141	53-146	
delta-BHC	ug/kg	9.9	12.6	128	16-140	
Dieldrin	ug/kg	19.7	25.5	129	53-161	
Endosulfan I	ug/kg	9.9	12.8	130	53-151	
Endosulfan II	ug/kg	19.7	27.2	138	52-154	
Endosulfan sulfate	ug/kg	19.7	23.5	119	50-147	
Endrin	ug/kg	19.7	26.5	134	52-168	
Endrin aldehyde	ug/kg	19.7	24.2	123	49-154	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

LABORATORY CONTROL SAMPLE: 3073073

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	9.9	13.0	132	51-157	
gamma-Chlordane	ug/kg	9.9	13.8	140	59-155	
Heptachlor	ug/kg	9.9	13.6	138	51-153	
Heptachlor epoxide	ug/kg	9.9	12.1	123	52-149	
Decachlorobiphenyl (S)	%.			85	10-141	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073074 3073075

Parameter	Units	60395082009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	21.1	20.9	33.2	31.4	157	150	16-166	5	20	
4,4'-DDE	ug/kg	ND	21.1	20.9	32.8	30.4	155	146	12-166	8	20	
4,4'-DDT	ug/kg	ND	21.1	20.9	30.2	28.1	143	134	10-191	7	20	
Aldrin	ug/kg	ND	10.6	10.5	16.2	15.6	153	149	10-161	4	20	
alpha-BHC	ug/kg	ND	10.6	10.5	17.8	16.4	168	157	15-162	8	20	M1
alpha-Chlordane	ug/kg	ND	10.6	10.5	15.6	14.8	147	142	14-156	5	20	
beta-BHC	ug/kg	ND	10.6	10.5	16.1	15.4	152	147	10-162	5	20	
delta-BHC	ug/kg	ND	10.6	10.5	15.4	14.7	146	141	10-157	5	20	
Dieldrin	ug/kg	ND	21.1	20.9	32.0	29.5	151	141	13-167	8	20	
Endosulfan I	ug/kg	ND	10.6	10.5	15.7	15.0	149	143	10-154	5	20	
Endosulfan II	ug/kg	ND	21.1	20.9	32.9	31.3	156	150	14-151	5	20	M1
Endosulfan sulfate	ug/kg	ND	21.1	20.9	27.4	25.8	130	123	10-148	6	20	
Endrin	ug/kg	ND	21.1	20.9	32.7	30.9	155	148	10-169	6	20	
Endrin aldehyde	ug/kg	ND	21.1	20.9	28.5	26.9	135	129	10-163	6	20	
gamma-BHC (Lindane)	ug/kg	ND	10.6	10.5	17.0	16.0	161	153	10-172	6	20	
gamma-Chlordane	ug/kg	ND	10.6	10.5	16.2	14.9	154	142	10-171	8	20	
Heptachlor	ug/kg	ND	10.6	10.5	16.2	15.4	153	147	10-173	5	20	
Heptachlor epoxide	ug/kg	ND	10.6	10.5	15.5	14.3	147	136	11-155	8	20	
Decachlorobiphenyl (S)	%.						88	89	10-141			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667217	Analysis Method: EPA 8081
QC Batch Method: EPA 3510	Analysis Description: 8081 GCS TCLP Pesticides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3072860 Matrix: Water

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	03/21/22 18:06	
Endrin	mg/L	<0.00010	0.00010	03/21/22 18:06	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	03/21/22 18:06	
Heptachlor	mg/L	<0.000050	0.000050	03/21/22 18:06	
Heptachlor epoxide	mg/L	<0.000050	0.000050	03/21/22 18:06	
Methoxychlor	mg/L	<0.00050	0.00050	03/21/22 18:06	
Toxaphene	mg/L	<0.0010	0.0010	03/21/22 18:06	
Decachlorobiphenyl (S)	%	63	10-102	03/21/22 18:06	

LABORATORY CONTROL SAMPLE: 3072861

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0027	137	57-155	
gamma-BHC (Lindane)	mg/L	0.001	0.0012	115	47-145	
Heptachlor	mg/L	0.001	0.00072	72	29-139	
Heptachlor epoxide	mg/L	0.001	0.0012	117	52-141	
Methoxychlor	mg/L	0.01	0.014	144	50-162	
Decachlorobiphenyl (S)	%			66	10-102	

MATRIX SPIKE SAMPLE: 3072862

Parameter	Units	60394386001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	ND	0.01	0.015	150	54-146	M1
gamma-BHC (Lindane)	mg/L	ND	0.005	0.0069	137	43-143	
Heptachlor	mg/L	ND	0.005	0.0051	102	17-140	
Heptachlor epoxide	mg/L	ND	0.005	0.0065	130	47-143	
Methoxychlor	mg/L	ND	0.05	0.075	150	51-153	
Decachlorobiphenyl (S)	%				70	10-102	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 667333

QC Batch Method: EPA 3546

Analysis Method: EPA 8082

Analysis Description: 8082 PCB Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073177

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1221 (Aroclor 1221)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1232 (Aroclor 1232)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1242 (Aroclor 1242)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1248 (Aroclor 1248)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1254 (Aroclor 1254)	ug/kg	<97.4	97.4	03/22/22 16:51	
PCB-1260 (Aroclor 1260)	ug/kg	<97.4	97.4	03/22/22 16:51	
Tetrachloro-m-xylene (S)	%	83	35-106	03/22/22 16:51	

LABORATORY CONTROL SAMPLE: 3073178

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	331	284	86	56-112	
PCB-1260 (Aroclor 1260)	ug/kg	331	270	82	48-119	
Tetrachloro-m-xylene (S)	%			83	35-106	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073179 3073180

Parameter	Units	50311718001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	325	327	160	188	49	58	10-143	16	20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	325	327	162	193	50	59	10-140	17	20	
Tetrachloro-m-xylene (S)	%						46	62	35-106			

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 667215

QC Batch Method: EPA 8151

Analysis Method: EPA 8151A

Analysis Description: 8151 GCS TCLP Herbicides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3072855

Matrix: Water

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	03/21/22 18:00	
2,4-D	mg/L	<0.0010	0.0010	03/21/22 18:00	
2,4-DCAA (S)	%.	97	10-142	03/21/22 18:00	

LABORATORY CONTROL SAMPLE: 3072856

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0048	96	48-139	
2,4-D	mg/L	0.005	0.0049	97	41-119	
2,4-DCAA (S)	%.			94	10-142	

MATRIX SPIKE SAMPLE: 3072857

Parameter	Units	50310808001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0050	0.025	0.026	103	10-154	
2,4-D	mg/L	<0.0050	0.025	0.027	109	10-142	
2,4-DCAA (S)	%.				88	10-142	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667289	Analysis Method: EPA 8270
QC Batch Method: EPA 3546	Analysis Description: 8270 Solid MSSV Microwave Short Spike
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073062 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<329	329	03/21/22 17:19	
2,4,5-Trichlorophenol	ug/kg	<329	329	03/21/22 17:19	
2,4,6-Trichlorophenol	ug/kg	<329	329	03/21/22 17:19	
2,4-Dichlorophenol	ug/kg	<329	329	03/21/22 17:19	
2,4-Dimethylphenol	ug/kg	<329	329	03/21/22 17:19	
2,4-Dinitrophenol	ug/kg	<1590	1590	03/21/22 17:19	
2,4-Dinitrotoluene	ug/kg	<329	329	03/21/22 17:19	
2,6-Dinitrotoluene	ug/kg	<329	329	03/21/22 17:19	
2-Chloronaphthalene	ug/kg	<329	329	03/21/22 17:19	
2-Chlorophenol	ug/kg	<329	329	03/21/22 17:19	
2-Methylnaphthalene	ug/kg	<329	329	03/21/22 17:19	
2-Methylphenol(o-Cresol)	ug/kg	<329	329	03/21/22 17:19	
2-Nitroaniline	ug/kg	<329	329	03/21/22 17:19	
2-Nitrophenol	ug/kg	<329	329	03/21/22 17:19	
3&4-Methylphenol(m&p Cresol)	ug/kg	<658	658	03/21/22 17:19	
3,3'-Dichlorobenzidine	ug/kg	<658	658	03/21/22 17:19	
3-Nitroaniline	ug/kg	<329	329	03/21/22 17:19	
4,6-Dinitro-2-methylphenol	ug/kg	<658	658	03/21/22 17:19	
4-Bromophenylphenyl ether	ug/kg	<329	329	03/21/22 17:19	
4-Chloro-3-methylphenol	ug/kg	<658	658	03/21/22 17:19	
4-Chloroaniline	ug/kg	<658	658	03/21/22 17:19	
4-Chlorophenylphenyl ether	ug/kg	<329	329	03/21/22 17:19	
4-Nitroaniline	ug/kg	<329	329	03/21/22 17:19	
4-Nitrophenol	ug/kg	<1590	1590	03/21/22 17:19	
Acenaphthene	ug/kg	<329	329	03/21/22 17:19	
Acenaphthylene	ug/kg	<329	329	03/21/22 17:19	
Anthracene	ug/kg	<329	329	03/21/22 17:19	
Benzo(a)anthracene	ug/kg	<329	329	03/21/22 17:19	
Benzo(a)pyrene	ug/kg	<329	329	03/21/22 17:19	
Benzo(b)fluoranthene	ug/kg	<329	329	03/21/22 17:19	
Benzo(g,h,i)perylene	ug/kg	<329	329	03/21/22 17:19	
Benzo(k)fluoranthene	ug/kg	<329	329	03/21/22 17:19	
Benzyl alcohol	ug/kg	<658	658	03/21/22 17:19	
bis(2-Chloroethoxy)methane	ug/kg	<329	329	03/21/22 17:19	
bis(2-Chloroethyl) ether	ug/kg	<329	329	03/21/22 17:19	
bis(2-Ethylhexyl)phthalate	ug/kg	<329	329	03/21/22 17:19	
bis(2chloro1methylethyl) ether	ug/kg	<329	329	03/21/22 17:19	
Butylbenzylphthalate	ug/kg	<329	329	03/21/22 17:19	
Chrysene	ug/kg	<329	329	03/21/22 17:19	
Di-n-butylphthalate	ug/kg	<329	329	03/21/22 17:19	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

METHOD BLANK: 3073062

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<329	329	03/21/22 17:19	
Dibenz(a,h)anthracene	ug/kg	<329	329	03/21/22 17:19	
Dibenzofuran	ug/kg	<329	329	03/21/22 17:19	
Diethylphthalate	ug/kg	<329	329	03/21/22 17:19	
Dimethylphthalate	ug/kg	<329	329	03/21/22 17:19	
Fluoranthene	ug/kg	<329	329	03/21/22 17:19	
Fluorene	ug/kg	<329	329	03/21/22 17:19	
Hexachloro-1,3-butadiene	ug/kg	<329	329	03/21/22 17:19	
Hexachlorobenzene	ug/kg	<329	329	03/21/22 17:19	
Hexachlorocyclopentadiene	ug/kg	<329	329	03/21/22 17:19	
Hexachloroethane	ug/kg	<329	329	03/21/22 17:19	
Indeno(1,2,3-cd)pyrene	ug/kg	<329	329	03/21/22 17:19	
Isophorone	ug/kg	<329	329	03/21/22 17:19	
N-Nitroso-di-n-propylamine	ug/kg	<329	329	03/21/22 17:19	
N-Nitrosodiphenylamine	ug/kg	<329	329	03/21/22 17:19	
Naphthalene	ug/kg	<329	329	03/21/22 17:19	
Nitrobenzene	ug/kg	<329	329	03/21/22 17:19	
Pentachlorophenol	ug/kg	<1590	1590	03/21/22 17:19	
Phenanthrene	ug/kg	<329	329	03/21/22 17:19	
Phenol	ug/kg	<329	329	03/21/22 17:19	
Pyrene	ug/kg	<329	329	03/21/22 17:19	
2,4,6-Tribromophenol (S)	%	73	20-121	03/21/22 17:19	
2-Fluorobiphenyl (S)	%	80	35-96	03/21/22 17:19	
2-Fluorophenol (S)	%	81	33-111	03/21/22 17:19	
Nitrobenzene-d5 (S)	%	89	32-105	03/21/22 17:19	
p-Terphenyl-d14 (S)	%	78	31-145	03/21/22 17:19	
Phenol-d5 (S)	%	88	35-114	03/21/22 17:19	

LABORATORY CONTROL SAMPLE: 3073063

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1660	1320	80	48-116	
2-Chlorophenol	ug/kg	1660	1200	72	58-100	
2-Methylnaphthalene	ug/kg	1660	1340	81	53-99	
4-Chloro-3-methylphenol	ug/kg	1660	1560	94	57-112	
4-Nitrophenol	ug/kg	1660	<1590	73	36-133	
Acenaphthene	ug/kg	1660	1310	79	61-98	
Acenaphthylene	ug/kg	1660	1250	76	61-98	
Anthracene	ug/kg	1660	1270	77	62-100	
Benzo(a)anthracene	ug/kg	1660	1280	77	64-101	
Benzo(a)pyrene	ug/kg	1660	1220	74	60-104	
Benzo(b)fluoranthene	ug/kg	1660	1330	80	65-107	
Benzo(g,h,i)perylene	ug/kg	1660	1370	83	61-108	
Benzo(k)fluoranthene	ug/kg	1660	1420	86	61-109	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

LABORATORY CONTROL SAMPLE: 3073063

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1660	1370	83	64-101	
Dibenz(a,h)anthracene	ug/kg	1660	1340	81	65-106	
Fluoranthene	ug/kg	1660	1310	79	63-111	
Fluorene	ug/kg	1660	1360	82	64-100	
Indeno(1,2,3-cd)pyrene	ug/kg	1660	1370	83	64-106	
N-Nitroso-di-n-propylamine	ug/kg	1660	1420	86	48-101	
Naphthalene	ug/kg	1660	1270	77	58-93	
Pentachlorophenol	ug/kg	1660	<1590	65	33-117	
Phenanthrene	ug/kg	1660	1300	78	62-102	
Phenol	ug/kg	1660	1180	71	56-101	
Pyrene	ug/kg	1660	1310	79	60-105	
2,4,6-Tribromophenol (S)	%			68	20-121	
2-Fluorobiphenyl (S)	%			72	35-96	
2-Fluorophenol (S)	%			72	33-111	
Nitrobenzene-d5 (S)	%			81	32-105	
p-Terphenyl-d14 (S)	%			71	31-145	
Phenol-d5 (S)	%			82	35-114	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073064 3073065

Parameter	Units	50311009002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	2040	2000	1470	1370	72	68	13-119	7	20	
2-Chlorophenol	ug/kg	ND	2040	2000	1450	1360	71	68	16-116	6	20	
2-Methylnaphthalene	ug/kg	ND	2040	2000	1670	1600	82	80	19-120	5	20	
4-Chloro-3-methylphenol	ug/kg	ND	2040	2000	1840	1710	91	85	22-124	7	20	
4-Nitrophenol	ug/kg	ND	2040	2000	<1950	<1920	72	75	10-139		20	
Acenaphthene	ug/kg	ND	2040	2000	1600	1600	78	80	25-114	0	20	
Acenaphthylene	ug/kg	ND	2040	2000	1510	1390	74	70	21-116	8	20	
Anthracene	ug/kg	ND	2040	2000	1530	1610	75	81	23-116	5	20	
Benzo(a)anthracene	ug/kg	ND	2040	2000	1750	1940	74	85	14-128	11	20	
Benzo(a)pyrene	ug/kg	ND	2040	2000	1580	1730	65	74	12-127	9	20	
Benzo(b)fluoranthene	ug/kg	ND	2040	2000	1820	2240	71	94	10-142	21	20	R1
Benzo(g,h,i)perylene	ug/kg	ND	2040	2000	1620	1650	80	83	16-120	2	20	
Benzo(k)fluoranthene	ug/kg	ND	2040	2000	1750	1580	86	79	15-131	10	20	
Chrysene	ug/kg	ND	2040	2000	1800	2010	74	86	11-132	11	20	
Dibenz(a,h)anthracene	ug/kg	ND	2040	2000	1470	1370	72	69	21-117	7	20	
Fluoranthene	ug/kg	0.55 mg/kg	2040	2000	2100	2800	76	112	10-143	29	20	R1
Fluorene	ug/kg	ND	2040	2000	1610	1600	79	80	18-122	1	20	
Indeno(1,2,3-cd)pyrene	ug/kg	ND	2040	2000	1620	1620	79	81	19-120	0	20	
N-Nitroso-di-n-propylamine	ug/kg	ND	2040	2000	1720	1610	84	80	24-109	7	20	
Naphthalene	ug/kg	ND	2040	2000	1610	1610	79	80	22-112	0	20	
Pentachlorophenol	ug/kg	ND	2040	2000	<1950	<1920	68	63	10-123		20	
Phenanthrene	ug/kg	ND	2040	2000	1820	2760	74	122	10-136	41	20	R1

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073064 3073065												
Parameter	Units	50311009002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Phenol	ug/kg	ND	2040	2000	1440	1410	71	70	14-122	2	20	
Pyrene	ug/kg	0.49 mg/kg	2040	2000	2050	2770	77	114	10-144	30	20	R1
2,4,6-Tribromophenol (S)	%						65	60	20-121			
2-Fluorobiphenyl (S)	%						71	67	35-96			
2-Fluorophenol (S)	%						69	68	33-111			
Nitrobenzene-d5 (S)	%						81	76	32-105			
p-Terphenyl-d14 (S)	%						63	58	31-145			
Phenol-d5 (S)	%						79	78	35-114			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667276	Analysis Method: EPA 8270
QC Batch Method: EPA 3510	Analysis Description: 8270 TCLP MSSV
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073018 Matrix: Water

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	03/21/22 16:48	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	03/21/22 16:48	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	03/21/22 16:48	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	03/21/22 16:48	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	03/21/22 16:48	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	03/21/22 16:48	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	03/21/22 16:48	
Hexachlorobenzene	mg/L	<0.010	0.010	03/21/22 16:48	
Hexachloroethane	mg/L	<0.010	0.010	03/21/22 16:48	
Nitrobenzene	mg/L	<0.010	0.010	03/21/22 16:48	
Pentachlorophenol	mg/L	<0.050	0.050	03/21/22 16:48	
Pyridine	mg/L	<0.010	0.010	03/21/22 16:48	
2,4,6-Tribromophenol (S)	%	66	47-127	03/21/22 16:48	
2-Fluorobiphenyl (S)	%	58	35-102	03/21/22 16:48	
2-Fluorophenol (S)	%	36	21-74	03/21/22 16:48	
Nitrobenzene-d5 (S)	%	74	40-115	03/21/22 16:48	
p-Terphenyl-d14 (S)	%	73	42-156	03/21/22 16:48	
Phenol-d5 (S)	%	25	15-48	03/21/22 16:48	

LABORATORY CONTROL SAMPLE: 3073019

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.029	58	30-85	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	76	52-117	
2,4,6-Trichlorophenol	mg/L	0.05	0.036	71	52-114	
2,4-Dinitrotoluene	mg/L	0.05	0.041	82	58-107	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.032	64	40-95	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.058	58	37-89	
Hexachloro-1,3-butadiene	mg/L	0.05	0.027	54	22-78	
Hexachlorobenzene	mg/L	0.05	0.034	68	46-79	
Hexachloroethane	mg/L	0.05	0.029	57	17-85	
Nitrobenzene	mg/L	0.05	0.046	93	50-110	
Pentachlorophenol	mg/L	0.05	<0.050	68	32-126	
Pyridine	mg/L	0.05	<0.010	16	18-69	L2
2,4,6-Tribromophenol (S)	%			72	47-127	
2-Fluorobiphenyl (S)	%			65	35-102	
2-Fluorophenol (S)	%			42	21-74	
Nitrobenzene-d5 (S)	%			85	40-115	
p-Terphenyl-d14 (S)	%			75	42-156	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

LABORATORY CONTROL SAMPLE: 3073019

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			28	15-48	

MATRIX SPIKE SAMPLE: 3073020

Parameter	Units	50310984001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.26	52	24-81	
2,4,5-Trichlorophenol	mg/L	ND	0.5	<0.50	82	34-129	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.37	74	33-123	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.42	84	35-116	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.29	59	24-102	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.54	54	18-99	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.25	50	15-79	
Hexachlorobenzene	mg/L	ND	0.5	0.34	68	29-86	
Hexachloroethane	mg/L	ND	0.5	0.26	53	14-79	
Nitrobenzene	mg/L	ND	0.5	0.43	85	27-117	
Pentachlorophenol	mg/L	ND	0.5	<0.50	77	15-151	
Pyridine	mg/L	ND	0.5	0.14	29	12-75	
2,4,6-Tribromophenol (S)	%.				77	47-127	
2-Fluorobiphenyl (S)	%.				70	35-102	
2-Fluorophenol (S)	%.				38	21-74	
Nitrobenzene-d5 (S)	%.				80	40-115	
p-Terphenyl-d14 (S)	%.				51	42-156	
Phenol-d5 (S)	%.				28	15-48	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch:	665755	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	Dry Weight/Percent Moisture
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

SAMPLE DUPLICATE: 3065485

Parameter	Units	5290701001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	99.5	99.5	0	5	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 665757

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

SAMPLE DUPLICATE: 3065493

Parameter	Units	5290701001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	0.49	0.49	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 665759

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: 2540G Total Volatile Solids

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

SAMPLE DUPLICATE: 3065498

Parameter	Units	5290701001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	76.5	76.3	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 665767	Analysis Method: EPA 9038
QC Batch Method: EPA 9038	Analysis Description: 9038 Sulfate Soil
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3065529 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<99.7	99.7	03/11/22 10:25	N2

LABORATORY CONTROL SAMPLE: 3065530

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	199	186	93	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3065531 3065532

Parameter	Units	50310766001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	1440	2120	2120	3420	3510	93	98	90-110	3	20	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667083	Analysis Method: EPA 9045
QC Batch Method: EPA 9045	Analysis Description: 9045 pH
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

SAMPLE DUPLICATE: 3071658

Parameter	Units	50311282001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	10.3	10.3	0	2	H3,PO

SAMPLE DUPLICATE: 3071659

Parameter	Units	50311054008 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.5	8.5	1	2	H3

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch:	667021	Analysis Method:	EPA 351.2
QC Batch Method:	EPA 351.2	Analysis Description:	351.2 TKN
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3071400 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.8	49.8	03/22/22 17:46	N2

LABORATORY CONTROL SAMPLE: 3071401

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	495	481	97	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3071402 3071403

Parameter	Units	50310808001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	98900	31800	31600	135000	130000	112	97	90-110	4	20	M0, N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 665739

Analysis Method: EPA 353.2

QC Batch Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3065360

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<4.9	4.9	03/10/22 14:30	N2
Nitrogen, NO2 plus NO3	mg/kg	<4.9	4.9	03/10/22 14:30	N2

LABORATORY CONTROL SAMPLE: 3065361

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	9.8	9.6	99	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.6	20.3	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3065362 3065363

Parameter	Units	50310887001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	72.7	72.7	74.3	73.6	101	100	90-110	1	20	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	146	146	157	156	107	106	90-110	1	20	N2

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50310808

QC Batch: 666096

Analysis Method: SM 4500-NH3 G

QC Batch Method: SM 4500-NH3 B

Analysis Description: 4500 Ammonia, Distilled

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3067484

Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	03/14/22 13:18	N2

LABORATORY CONTROL SAMPLE: 3067485

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.8	21.5	108	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3067486 3067487

Parameter	Units	50310887001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	243	148	148	404	397	108	104	90-110	2	20	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667186	Analysis Method: SM 4500-Cl-E
QC Batch Method: SM 4500-Cl-E	Analysis Description: 4500 Chloride
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3072534 Matrix: Solid

Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<97.8	97.8	03/20/22 12:32	N2

LABORATORY CONTROL SAMPLE: 3072535

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	196	191	98	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3072536 3072537

Parameter	Units	50310766001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<26100	52200	52200	51200	52100	97	99	90-110	2	20	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50310808

QC Batch: 667457	Analysis Method: SM 4500-CN-E
QC Batch Method: SM 4500-CN-C	Analysis Description: 4500CNE Cyanide
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50310808001

METHOD BLANK: 3073533 Matrix: Solid
Associated Lab Samples: 50310808001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Cyanide	mg/kg	<0.49	0.49	03/22/22 18:00	N2

LABORATORY CONTROL SAMPLE: 3073534

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	4.9	5.1	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3073535 3073536

Parameter	Units	50310808001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	<31.8	316	316	311	296	97	93	90-110	5	20	N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50310808

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	A matrix spike/matrix spike duplicate was not performed for this batch due to insufficient sample volume.
D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H3	Sample was received or analysis requested beyond the recognized method holding time.
H7	Re-extraction or re-analysis could not be performed within method holding time.
L2	Analyte recovery in the laboratory control sample (LCS) was below QC limits. Results for this analyte in associated samples may be biased low.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
PO	The reported result is outside the range of the pH buffer solutions used to check the calibration of the pH meter.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50310808

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50310808001	Digester #4	EPA 3546	667291	EPA 8081	667396
50310808001	Digester #4	EPA 3510	667217	EPA 8081	667341
50310808001	Digester #4	EPA 3546	667333	EPA 8082	667497
50310808001	Digester #4	EPA 8151	667215	EPA 8151A	667375
50310808001	Digester #4	EPA 3050	667371	EPA 6010	667702
50310808001	Digester #4	EPA 3010	667082	EPA 6010	667433
50310808001	Digester #4	EPA 3050B	666526	EPA 6020	667055
50310808001	Digester #4	EPA 7470	667101	EPA 7470	667250
50310808001	Digester #4	EPA 7471	667251	EPA 7471	667351
50310808001	Digester #4	EPA 3546	667289	EPA 8270	667384
50310808001	Digester #4	EPA 3510	667276	EPA 8270	667382
50310808001	Digester #4	EPA 8260	667532		
50310808001	Digester #4	EPA 5030/8260	667363		
50310808001	Digester #4	SM 2540G	665755		
50310808001	Digester #4	SM 2540G	665757		
50310808001	Digester #4	SM 2540G	665759		
50310808001	Digester #4	EPA 9038	665767	EPA 9038	665779
50310808001	Digester #4	EPA 9045	667083		
50310808001	Digester #4	EPA 351.2	667021	EPA 351.2	667594
50310808001	Digester #4	EPA 353.2	665739	EPA 353.2	665744
50310808001	Digester #4	SM 4500-NH3 B	666096	SM 4500-NH3 G	666184
50310808001	Digester #4	SM 4500-CI-E	667186	SM 4500-CI-E	667188
50310808001	Digester #4	SM 4500-CN-C	667457	SM 4500-CN-E	667579

REPORT OF LABORATORY ANALYSIS

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WO#: 50310808



50310808

Required Client Information:		Required Project Information:		Invoice Information:	
Company:	City of Cadillac	Report To:	Cindy Tomaszewski	Attention:	
Address:	1121 Plett Rd	Copy To:		Company Name:	
	Cadillac, MI 49601			Address:	
Email:	domaszewski@cadillac-mi.net	Purchase Order #:		Pace Quote:	
Phone:	231-775-2368	Project Name:	Biosolids	Pace Project Manager:	brian.hall@pacelabs.com,
		Project #:		Pace Profile #:	8311
Requested Due Date:					
				Regulatory Agency	
				State / Location	
				MI	

[illegible][illegible]

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>3.9.22</u>		Evaluated by: <u>WDC</u>		WO# : 50310808 PM: BJH Due Date: 03/23/22 CLIENT: GR-Cadillac	
Client: <u>City of Cadillac</u>		Profile ID:			
Project Manager: <u>BJH</u>		Due Date:			
Rush TAT Requested: <u>YES</u> <u>NO</u>		Due Date:		Non Conformance Form Required: <u>YES</u> <u>NO</u>	
Lab Notified of Rush or Short Holds: <u>YES</u> <u>NO</u>					
Samples Received Via: <u>FedEx</u> <u>UPS</u> <u>Client</u> <u>Pace Courier</u> Other: _____					Comments:
Custody Seals Present and Intact:		YES	NO	<u>N/A</u>	
Received Sample Information Form(s): Drinking Waters Only		YES	NO	<u>N/A</u>	
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)		YES	NO	<u>N/A</u>	
Short Holds Present (< 72 Hours):		YES	NO	<u>N/A</u>	
Samples Received in Hold:		YES	NO	<u>N/A</u>	
Custody Signatures Present:		YES	NO	<u>N/A</u>	
Collector Signature Present:		YES	NO	<u>N/A</u>	
Packing Material Used:		YES	NO	<u>N/A</u>	
Samples Collected Today and On Ice:		YES	NO	<u>N/A</u>	
IR Gun #: 280 <u>281</u>		Digital Thermometer #: 282 283			
Ice Type: <u>WET Bagged / WET Loose</u> <u>BLUE</u> <u>NONE</u>		1. Cooler Temp Upon Receipt: <u>1.9/0.9</u> °C			
Ice Location: <u>TOP</u> <u>BOTTOM</u> <u>MIDDLE</u> <u>DISPERSED</u>		Temp should be 0-6°C (Initial/Corrected)			
Temp Blank Received:		YES	NO	<u>N/A</u>	
Containers Intact:		YES	NO	<u>N/A</u>	
Correct Containers:		YES	NO	<u>N/A</u>	
Sufficient Volume:		YES	NO	<u>N/A</u>	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: _____ <i>Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl</i>		YES	NO	<u>N/A</u>	
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ (SVOC/Pest 625, PCB 608, Total/Amenable Cyanide)		YES	NO	<u>N/A</u>	
VOA Headspace Acceptable (<6mm):		YES	NO	<u>N/A</u>	
Trip Blank Received: <u>HCl</u> <u>MeOH</u> <u>TSP</u> <u>OTHER</u>		YES	NO	<u>N/A</u>	
Comments:		2. Cooler Temp Upon Receipt: _____ °C			
		3. Cooler Temp Upon Receipt: _____ °C			
		4. Cooler Temp Upon Receipt: _____ °C			